Book-campaign-performance-report

Benkly

2026-03-23

Introduction

This report acts as a follow-on to the report titled `Book-sales-report`, which deals with a scenario where a data analyst has been tasked to provide a high-level financial performance report by a company which publishes textbooks related to learning the R programming language.

For this report, we imagine the company has provided additional sales information for the fiscal year of 2019. The company launched a campaign on July 1st 2019 to promote their books and would like to know whether their is evidence to suggest the campaign was successful at increasing sales and improving review quality.

Aims

- To summarize sales and review sentiment before and after the campaign start date to determine whether there is evidence to suggest the campaign was a success.

To do this, date/time manipulation and string manipulation methods are necessary.

Dataset

The dataset was sourced from Dataquest.

Data Dictionary

Column	Description
date	Date of sale
user_submitted_review	Text based review from the buyer
title	The title of the purchased book
total_purchased	The quantity of books purchased
customer_type	Whether the customer is an Individual or Business

Loading Libraries and Importing the Data

```r
# Loading libraries
library(conflicted)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
```

```r
library(lubridate)

# Resolve filter conflict
conflicts_prefer(dplyr::filter)
```

```
## [conflicted] Will prefer dplyr::filter over any other package.
```

```r
# Reading the Data
sales_2019 <- read_csv("sales2019.csv")
```

```
## Rows: 5000 Columns: 5
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): date, user_submitted_review, title, customer_type
## dbl (1): total_purchased
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Exploring the Dataset

```r
# Dimension of the Dataframe
dim(sales_2019)
```

```
## [1] 5000    5
```

```r
# Column Names
cols <- colnames(sales_2019)
cols
```

```
## [1] "date"                  "user_submitted_review" "title"
## [4] "total_purchased"       "customer_type"
```

```r
# Global view of the data
glimpse(sales_2019)
```

```
## Rows: 5,000
## Columns: 5
## $ date                  <chr> "5/22/19", "11/16/19", "6/27/19", "11/06/2019", ~
## $ user_submitted_review <chr> "it was okay", "Awesome!", "Awesome!", "Awesome!~
## $ title                 <chr> "Secrets Of R For Advanced Students", "R For Dum~
## $ total_purchased       <dbl> 7, 3, 1, 3, NA, 1, 5, NA, 7, 1, 7, NA, 3, 2, 0, ~
## $ customer_type         <chr> "Business", "Business", "Individual", "Individua~
```

```r
# Missing Values
missing <- function(df) {
  missing_values <- rep(0, times=length(colnames(df)))
  names(missing_values) <- colnames(df)
  i <- 1

  for (col in colnames(df)) {
    missing_values[i] <- is.na(df[[col]]) %>% sum
    i <- i + 1
  }
  return(missing_values)
}

missing(sales_2019)
```

```
##                 date user_submitted_review                 title
##                    0                   885                     0
##      total_purchased         customer_type
##                  718                     0
```

The dataset contains 5000 records and 5 columns. There are 885 missing values in the `user_submitted_review` column and 718 missing values in the `total_purchased` column.

Since one of our primary concerns is with sales performance before and after the campaign date, we ideally do not want to discard records with missing data in the `total_purchased` field. We can instead impute the missing data with the mean as a best estimate for the missing values.

In the case of the missing values in the `user_submitted_review` field, we can discard any records with no review as we cannot impute this data.

Handling Missing Values

```r
# Remove all records with no user review

sales_2019_clean <- sales_2019 %>%
  filter(!is.na(user_submitted_review))

# Calculate imputation value

average_purchase_amount <- sales_2019_clean %>%
  filter(!is.na(total_purchased)) %>%
  pull(total_purchased) %>%
  mean

# Impute missing purchase amounts with the calculated value

sales_2019_clean <- sales_2019_clean %>%
  mutate(
    imputed_purchases = if_else(is.na(total_purchased), round(average_purchase_amount, 1),
```

```
                                total_purchased)
  )

# Check for missing values on the cleaned dataset
missing(sales_2019_clean)
```

```
##                  date user_submitted_review                 title
##                     0                     0                     0
##       total_purchased         customer_type     imputed_purchases
##                   583                     0                     0
```

```
# Dimensions of cleaned dataset
dim(sales_2019_clean)
```

```
## [1] 4115    6
```

From the original 5000 records, 885 records (17.7%) have been removed corresponding to missing user review fields, leaving 4115 records. The remaining 583 null values in the `total_purchased` field have been imputed using the average of the `total_purchased` field (with nulls removed).

Handling the Review Column

The `user_submitted_review` field is more complicated in that there is no specific format that the data must fit into. It is a free-form text field which users can input any sequence of ASCII characters. For this report, we would like to assign a sentiment to each review as either positive, negative or neutral.

```
unique_reviews <- sales_2019_clean %>%
  pull(user_submitted_review) %>%
  unique()

unique_reviews
```

```
## [1] "it was okay"
## [2] "Awesome!"
## [3] "Hated it"
## [4] "Never read a better book"
## [5] "OK"
## [6] "The author's other books were better"
## [7] "A lot of material was not needed"
## [8] "Would not recommend"
## [9] "I learned a lot"
```

It seems there are only a handful of unique reviews (this is a simplified case). Since there are a relatively small number of unique reviews, we can assign sentiment to each using a custom function.

```
sentiment <- function(review) {
  review <- str_to_lower(review)
  sentiment_class <- case_when(
    str_detect(review, "awesome") ~ "Positive",
    str_detect(review, "learned a lot") ~ "Positive",
```

```r
    str_detect(review, "never") ~ "Positive",
    str_detect(review, "not recommend") ~ "Negative",
    str_detect(review, "not needed") ~ "Negative",
    str_detect(review, "hate") ~ "Negative",
    str_detect(review, "other books were better") ~ "Negative",
    TRUE ~ "Neutral" # captures reviews which do not contain any of the substrings listed above
  )
}

sales_2019_clean <- sales_2019_clean %>%
  mutate(
    sentiment = unlist(map(user_submitted_review, sentiment)),
    sentiment_score = case_when(
      sentiment == "Positive" ~ 1,
      sentiment == "Neutral" ~ 0,
      TRUE ~ -1
    )
  )

knitr::kable(head(sales_2019_clean, 20))
```

date	user_submitted_review	title	total_purchased	customer_type	imputed_purchases	sentiment	sentiment_score
5/22/19	it was okay	Secrets Of R For Advanced Students	7	Business	7	Neutral	0
11/16/19	Awesome!	R For Dummies	3	Business	3	Positive	1
6/27/19	Awesome!	R For Dummies	1	Individual	1	Positive	1
11/06/2019	Awesome!	Fundamentals of R For Beginners	3	Individual	3	Positive	1
7/18/19	Hated it	Fundamentals of R For Beginners	NA	Business	4	Negative	-1
1/28/19	Never read a better book	Secrets Of R For Advanced Students	1	Business	1	Positive	1
2/20/19	Hated it	R For Dummies	5	Business	5	Negative	-1
12/17/19	Awesome!	R For Dummies	NA	Business	4	Positive	1
7/13/19	OK	R vs Python: An Essay	7	Business	7	Neutral	0
6/22/19	The author's other books were better	R For Dummies	1	Business	1	Negative	-1
3/14/19	A lot of material was not needed	R For Dummies	7	Business	7	Negative	-1
12/19/19	A lot of material was not needed	R For Dummies	NA	Business	4	Negative	-1
3/14/19	Awesome!	Fundamentals of R For Beginners	3	Business	3	Positive	1
9/23/19	Awesome!	R For Dummies	2	Business	2	Positive	1
1/17/19	it was okay	R For Dummies	6	Business	6	Neutral	0
04/08/2019	Would not recommend	R For Dummies	3	Business	3	Negative	-1
7/21/19	I learned a lot	R For Dummies	2	Individual	2	Positive	1
12/12/2019	Would not recommend	Fundamentals of R For Beginners	NA	Business	4	Negative	-1
3/14/19	it was okay	R For Dummies	4	Business	4	Neutral	0

date	user_submitted_review	title	total_purchased	customer_type	imputed_purchases	sentiment	sentiment_score
09/10/2019	Would not recommend	R For Dummies	4	Individual	4	Negative	-1

Preparing the `date` Field

Before moving to segmenting the dataset by date, we first need to prepare the date data for use. As it stands currently, the date is expressed as a string representation. The **lubridate** package can be used to format and parse datetime data.

```
campaign_date <- "07-01-2019" %>% mdy %>% as.numeric() # unix-time representation of the campaign date

sales_2019_clean <- sales_2019_clean %>%
  mutate(
  date = mdy(date),
  during_campaign = if_else(date >= campaign_date, TRUE, FALSE)
)

knitr::kable(head(sales_2019_clean))
```

date	user_submitted_review	title	total_purchased	customer_type	imputed_purchases	sentiment	sentiment_score	during_campaign
2019-05-22	it was okay	Secrets Of R For Advanced Students	7	Business	7	Neutral	0	FALSE
2019-11-16	Awesome!	R For Dummies	3	Business	3	Positive	1	TRUE
2019-06-27	Awesome!	R For Dummies	1	Individual	1	Positive	1	FALSE
2019-11-06	Awesome!	Fundamentals of R For Beginners	3	Individual	3	Positive	1	TRUE
2019-07-18	Hated it	Fundamentals of R For Beginners	NA	Business	4	Negative	-1	TRUE
2019-01-28	Never read a better book	Secrets Of R For Advanced Students	1	Business	1	Positive	1	FALSE

```
# Save cleaned data
write.csv(sales_2019_clean, "sales2019clean.csv", row.names=FALSE)
```

We have successfully categorized each record as either pre- or post-campaign, assigned a sentiment (and sentiment score) to each record and have a measure of sales performance using the `imputed_purchases` field. The data is ready for analysis.

Analysis

We can begin by aggregating the data pre- and post-campaign to get an insight into whether an improvement in sales and sentiment occurred.

```r
# create a mode function to find the statistical mode of the sentiment column
mode <- function(x) {
  unique_vals <- unique(x)
  unique_vals[which.max(tabulate(match(x, unique_vals)))]
}

campaign_performance_breakdown <- sales_2019_clean %>%
  group_by(during_campaign) %>%
  summarize(
    sentiment = mode(sentiment),
    avg_sentiment_score = round(mean(sentiment_score), 3),
    total_purchases = sum(imputed_purchases),
    avg_quantity_purchased_per_sale = round(mean(imputed_purchases), 2)
  )

knitr::kable(campaign_performance_breakdown)
```

during_campaign	sentiment	avg_sentiment_score	total_purchases	avg_quantity_purchased_per_sale
FALSE	Negative	-0.126	8215	4.01
TRUE	Negative	-0.123	8194	3.97

- There is only a very marginal improvement in review sentiment during the period of time when the campaign was running.
- The number of books purchased actually declined slightly during the campaign period.

These changes are small, so will certainly be statistically insignificant. Taking the data as a whole, there is insufficient evidence to support that the campaign was successful at improving sentiment or purchase of books.

It could be the case that some customers responded better to the campaign depending on whether they are an individual versus a business. We can sub-divide the data further to explore this:

```r
campaign_performance_breakdown_by_customer_type <- sales_2019_clean %>%
  group_by(during_campaign, customer_type) %>%
  summarize(.groups="drop_last",
    sentiment = mode(sentiment),
    avg_sentiment_score = round(mean(sentiment_score), 3),
    total_purchases = sum(imputed_purchases),
    avg_quantity_purchased_per_sale = round(mean(imputed_purchases), 2)
  ) %>%
  arrange(customer_type)

knitr::kable(campaign_performance_breakdown_by_customer_type)
```

during_campaign	customer_type	sentiment	avg_sentiment_score	total_purchases	avg_quantity_purchased_per_sale
FALSE	Business	Negative	-0.124	5615	4.00
TRUE	Business	Negative	-0.108	5745	3.96
FALSE	Individual	Negative	-0.130	2600	4.03
TRUE	Individual	Negative	-0.158	2449	3.99

- Businesses show a slight improvement in both sales and review sentiment comparing pre-campaign performance to post-campaign performance.
- There is a decline in review sentiment and sales for individual customers however.

Finally, we can breakdown the sentiment change and sales performance for specific titles:

```
campaign_performance_breakdown_by_customer_type <- sales_2019_clean %>%
  group_by(during_campaign, title) %>%
  summarize(.groups="drop_last",
    sentiment = mode(sentiment),
    avg_sentiment_score = round(mean(sentiment_score), 3),
    total_purchases = sum(imputed_purchases),
    avg_quantity_purchased_per_sale = round(mean(imputed_purchases), 2)
  ) %>%
  arrange(title)

knitr::kable(campaign_performance_breakdown_by_customer_type)
```

during_campaign	title	sentiment	avg_sentiment_score	total_purchases	avg_quantity_purchased_per_sale
FALSE	Fundamentals of R For Beginners	Negative	-0.168	3095	4.02
TRUE	Fundamentals of R For Beginners	Negative	-0.117	2834	3.91
FALSE	R For Dummies	Negative	-0.111	2627	3.99
TRUE	R For Dummies	Negative	-0.109	2780	4.03
FALSE	R Made Easy	Negative	-0.250	15	3.75
TRUE	R Made Easy	Neutral	0.000	24	4.80
FALSE	R vs Python: An Essay	Negative	-0.030	1271	4.18
TRUE	R vs Python: An Essay	Negative	-0.156	1173	3.81
FALSE	Secrets Of R For Advanced Students	Negative	-0.123	966	3.83
TRUE	Secrets Of R For Advanced Students	Negative	-0.168	1155	4.12
FALSE	Top 10 Mistakes R Beginners Make	Negative	-0.246	241	3.95
TRUE	Top 10 Mistakes R Beginners Make	Positive	0.034	228	3.93

We can draw some interesting conclusions from this breakdown:

- The general sentiment for `Fundamentals of R For Beginners` noticeably improved, however, the number of sales of the book declined.
- More copies of `R For Dummies` were sold post-campaign, though the review sentiments did not significantly change.
- Very few copies of `R Made Easy` sold during either period. A slight improvement in sentiment and sales is noted during the campaign, but due to the small sample size, should not be taken too seriously.

- `R vs Python: An Essay` performed considerably worse during the campaign period, both sentiment and sales declined.
- Sales of `Secrets Of R For Advanced Students` improved during the campaign, however, the sentiment of the reviews worsened.
- The book `Top 10 Mistakes R Beginners Make` is the second worst performing title sold by the company. We can note an improvement in review sentiment post-campaign, however, sales declined slightly.